

**Cooperation in Heterogeneous Networks:
The Role of Committed Agents and Network Topology in Public
Goods Games**

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Goods Games**

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BS-MS in ECONOMICS

by

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CERTIFICATE

This is to certify that the Bachelor Thesis Project – 1 entitled
“Cooperation in Heterogeneous Networks: The Role of Committed Agents and Network Topology in Public Goods Games”
submitted by **Parikh Ansh Ashokkumar** (Roll No.: 22HS23012) in partial fulfillment
of the requirements for the award of the degree of **BS-MS in ECONOMICS** at the
Indian Institute of Technology, Kharagpur, is a record of bona fide work carried
out by him under my supervision.

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LIST OF SYMBOLS

A_R	Set of rational agents
A_D	Set of duty-driven agents
f_D	Fraction of duty-driven agents
r	Synergy factor (public goods multiplier)
c	Cost of cooperation
T	Number of time steps
$\rho_R(t)$	Rational cooperation rate at time t
N_i	Neighbourhood of agent i
π_i	Payoff of agent i

ABSTRACT

Cooperation is a fundamental component of human societies, yet classical economic models predict its collapse under individual rationality. This thesis investigates how behavioural heterogeneity and network topology jointly determine cooperative outcomes in Public Goods Games. By developing an agent-based model with rational and duty-driven agents interacting across five network topologies—Complete, Small-World, Scale-Free, Hub–Rational, and Hub–Duty-Driven—this study explores the structural and positional mechanisms that sustain cooperation.

The findings reveal that cooperation cannot emerge in well-mixed populations but thrives in structured networks where local clustering or moral anchoring occurs. In particular, the strategic placement of duty-driven agents in highly connected nodes (hubs) substantially amplifies cooperation, even when they form a small fraction of the population. Conversely, rational hubs trigger systemic collapse through a "super-free-rider" effect. The results demonstrate that the architecture of influence, rather than agent proportion alone, determines collective outcomes. These insights contribute to the growing synthesis of behavioural and network science, suggesting that cooperation is not an individual virtue but a structural achievement.

Keywords: Cooperation, Public Goods Game, Network Topology, Agent-Based Model, Heterogeneous Agents, Social Networks, Collective Action

Chapter 1

Introduction

1.1 The Public Goods Dilemma

A public good is defined as a non-rivalrous and non-excludable good. Public goods such as clean air, roads, or open-source software are foundational for our collective welfare. The non-excludability of public goods creates a fundamental problem: each individual has an incentive to free-ride. The Public Goods Game (PGG) formalises this problem. In a group of N agents, each chooses whether to contribute a cost c to a public good. The total contribution is multiplied by a synergy factor r and divided equally among all agents, regardless of their contributions. When $1 < r < N$, the Nash equilibrium outcome is to defect, but the socially optimal outcome is to cooperate. Rational agents will defect, resulting in the under-provision of the public good or the free-riding problem [1].

Yet, in reality, we do not see societies always collapsing into selfishness. Cooperation emerges frequently even in large, anonymous settings. Even when there is no apparent benefit, we see people cooperating in environmental activism, volunteer computing, and open collaboration. Explaining this cooperation within the assumption of rationality is the central problem of evolutionary and behavioural economics.

1.2 The Puzzle of Real-World Cooperation

Many mechanisms, such as repeated interactions, punishment, or reputation effects [2] were employed to explain cooperation, but their assumptions, such as the homogeneity of the agents and fully mixed populations, are rarely met in real life. In reality, humans have different utility functions, which are functions of various parameters such as moral or social norms, which the rationality assumption doesn't take into account. Likewise, we get influenced by our family, friends, and networks rather than randomly. This makes our interactions part of a large social network of agents. These features fundamentally alter collective dynamics.

[3] and [4] have shown that structured populations can sustain cooperation. [5] have shown that a small number of committed agents can maintain and trigger large-scale cooperation. The joint role of these two mechanisms, heterogeneity and structure, remains underexplored.

1.3 A New Framework: Heterogeneity and Structure

This thesis integrates these insights into a unified model of cooperation in heterogeneous networks. Two complementary hypotheses guide the analysis:

- **Behavioural Hypothesis:** Populations consist of
 - Rational Agents (A_R) – adaptive, payoff-maximising individuals who revise their strategy based on success; and
 - Duty-Driven Agents (A_D) – unconditional cooperators who contribute regardless of payoff.

The fraction f_D of committed agents represents moral or normative commitment within the population.

- **Structural Hypothesis:** Interactions between agents happen in networks, not in well-mixed settings. The study considers three canonical topologies:
 - Complete Graph – every agent is connected to every other agent.
 - Small-World Network – each agent is connected to their neighbour and some other agents via shortcuts [6].
 - Scale-Free Network – central hub is connected to every agent [7].

These structures determine whether cooperative behaviour diffuses, stagnates, or collapses. Of special interest is the position of committed agents: when a moral actor occupies a central hub, their influence may stabilise cooperation far beyond their proportion.

1.4 Research Question and Objectives

The central question is:

How does network topology mediate the influence of a minority of committed agents on the cooperative behaviour of a rational majority?

From this follow three objectives:

1. Examine how cooperation evolves across different network topologies.
2. Identify how the fraction and positioning of committed agents (f_D) affect long-run cooperation.
3. Determine whether deterministic behavioural updating—copying the more successful neighbour—is sufficient to generate stable cooperation.

1.5 Thesis Outline

- Chapter 2 is a literature review.
- Chapter 3 presents the mathematical and computational model.
- Chapter 4 analyses the structural dependency for cooperation.
- Chapter 5 explores the placement dependency of the committed agents.
- Chapter 6 concludes with directions for further research.

In summary, this thesis argues that cooperation among agents is dependent upon various factors, such as the relative presence of duty-driven agents, their positions in a network structure, and the type of network structure.

Chapter 2

Literature Review

2.1 The Public Goods Game and the Logic of Collective Action

Our understanding of cooperation primarily comes from The Public Goods Game (PGG). PGG tries to capture the tension between individual benefit and the collective welfare of society. When there is no exclusion based on whether the agent contributes a cost c or not to the public fund, which is then multiplied by the synergy factor $r < N$ and distributed equally among all agents, the game will result in total defection [8]. The Nash equilibrium of full defection is the opposite of the social welfare maximisation requirement of full cooperation. Many experiments have shown that this prediction of PGG is not realised, and agents cooperate significantly [9, 10].

Several modifications, like repetition and punishment in the PGG, have been proposed to explain this behaviour of agents. Repetitions in the model can explain reciprocity, and punishment mechanisms have shown cooperation; a reputation system can promote trust [11]. These mechanisms still cannot explain the anonymous cooperation at a large scale, such as open-source development or environmental activism.

2.2 Behavioral Heterogeneity: Beyond *Homo Economicus*

Traditional models treat every agent as rational, i.e., the utility maximiser or the one who acts for his/her own personal benefits, regardless of social welfare. However, Behavioural economics and game theory have identified three types of agents that can explain cooperation:

- Conditional cooperators who contribute when others do.
- Free-riders who exploit others' efforts.
- Altruists or committed cooperators who contribute regardless of others' actions [12, 5].

Evolutionary game theory shows that the presence of a small fraction of rational agents in the mixture of both rational and irrational agents can make the entire population act rationally [13].

Moral norms or institutions persist even after their incentives change. We can explain this through irreversibility and path dependence in collective outcomes, which is caused by the heterogeneity of agents creating hysteresis effects.

2.3 Cooperation on Networks: The Structural Dimension

In the real world, the information is imperfect. Agents do not interact with the entire population, but the local structure of which they are a part. These phenomena can not be explained by heterogeneity; we need a network that models the overlapping, localised networks of relationships between agents. Many studies have shown that the structure of the network itself can promote certain prosocial behaviours [3, 4].

In a random or well-mixed network, defectors will always win, leading to under-provision of the public good. But there can be some structure in which cooperators

can protect themselves from exploitation. This can be examined by studying the topology of the network, i.e., how nodes are connected in the network.

Different network structures can be used:

- Regular lattices enable cooperators to survive by forming local clusters.
- Small-world networks [6] are like a lattice but with a few shortcuts that ensure the stability and spread of moral agents.
- Scale-free networks [7] are networks with a highly connected central node or hub. These hubs can act as amplifiers of cooperation.

Social media is a great example of a scale-free network where the influencers act as a central node with very high connectivity. They influence the norms of their viewers way more than other nodes. Empirical evidence supports this theoretical insight. A cooperative hub can trigger a cascade of imitation, while a selfish hub can lead to systemic collapse [14].

2.4 The Interaction of Heterogeneity and Structure: The Research Gap

Research shows that heterogeneous degree distribution can enhance cooperation [15], and a small committed minority can stabilise it [13]. A critical gap that is the intersection of both the behavioural science and network science is largely unexplored. Network-based models assume the homogeneity of agents, while behavioural models fail to capture the network effects.

My thesis tries to fill this gap by examining the behavioural heterogeneity among agents in the form of duty-driven vs rational agents in different network topologies, like Complete, Small-World, and Scale-Free, in the Public Good Game. The thesis will also examine whether the cooperation persists in the long run when rational

agents can adopt their neighbours' choice at every step of the game if the neighbours' payoff is higher than theirs.

2.5 Summary

The literature has shown that heterogeneity among agents and the network structure through which agents are connected are the keys to explaining cooperation. The following chapter will formalise an agent-based mathematical model, which integrates the heterogeneous behaviour and network structure within a unified evolutionary framework.

Chapter 3

Model and Methodology

3.1 Overview of the Agent-Based Model

“How heterogeneous agents and network topology jointly shape cooperation?” To answer this question, we make an extension of the classical evolutionary Public Good Game framework. The model incorporates two types of agents, namely rational agents and duty-driven agents, and their interactions across various network topologies. For this study, we are limiting ourselves to only endogenous cooperation arising from local interactions between agents; we will not consider any exogenously imposed mechanism.

We define that agent j is a neighbour of agent i if there exists an edge between them. At each time step, there is a local PGG for every agent with their neighbours. Every agent will get some payoff. And according to that payoff, rational agents will decide whether to update their strategy using a copy-if-better rule. By varying the proportion of duty-driven agents, the initial cooperative rational agents, the synergy factor, and the position of duty-driven and rational agents across various network topologies, we will study how structural and behavioural heterogeneity interact to determine the equilibrium.

3.2 Agents and Types

There are N agents in a population. Each agent is a member of set $A = \{1, 2, \dots, N\}$.

There are two types of agents, and they do not change their behaviour (whether they are rational or duty-driven) over time.

- **Rational Agents** (A_R) maximize their payoff; they can change their strategies (to cooperate or defect) over time based on their neighbours' payoff.
- **Duty-Driven Agents** (A_D) will always cooperate without any condition; their utility is derived from the act of contributing, acting as moral anchors in the system.

Formally,

$$A_R \cup A_D = A, \quad A_R \cap A_D = \emptyset$$

The fraction of duty-driven agents is denoted by:

$$f_D = \frac{|A_D|}{N}$$

Hence, the number of rational agents equals $|A_R| = N(1 - f_D)$. The parameter f_D becomes a key experimental variable, capturing the density of individuals driven by moral or duty considerations within the population.

3.3 Network Interaction Structure

Interactions occur over a network $G = (A, E)$, where A represents agents (nodes) and E represents edges (social connections). Each agent i interacts only with its neighbourhood $N_i \subseteq A$.

To explore the role of structure, five topologies are examined:

- **Complete Graph:** A well-mixed network where every agent interacts with every other agent. Degree $k_i = N - 1$.
- **Small-World Network (Watts–Strogatz):** Constructed from a ring lattice of average degree $k = 4$, with each edge rewired with probability $p = 0.3$. Combines local clustering with short global path lengths.
- **Scale-Free Network (Barabási–Albert):** Generated through preferential attachment. New nodes connect to existing ones with probability proportional to their degree. Captures real-world heterogeneity where few nodes (hubs) have many links.
- **Hub–Duty-Driven (Star Graph):** A star network where the central hub (node 0) is a duty-driven agent. Represents a moral leader anchoring a group of followers.
- **Hub–Rational (Star Graph):** A star network where the central hub is rational. Represents a powerful but self-interested individual influencing many.

The degree of an agent i is $k_i = |N_i|$. For complete networks, $N_i = A \setminus \{i\}$; for all others, N_i depends on network construction.

3.4 Game Dynamics: The Public Goods Game

At each time step t , every agent plays a PGG within its inclusive neighbourhood $N_i \cup \{i\}$. Each agent chooses a strategy:

$$s_i(t) \in \{0, 1\}, \quad 1 = \text{Cooperate (contribute)}, \quad 0 = \text{Defect (not contribute)}.$$

Duty-driven agents have a fixed cooperative strategy:

$$s_i(t) = 1, \quad \forall i \in A_D, \forall t$$

Rational agents are initialized such that $s_i(0) = 1$ with probability $\rho_R(0)$, representing the initial rational cooperation rate.

The total number of cooperators in agent i 's group is:

$$C_i(t) = s_i(t) + \sum_{j \in N_i} s_j(t)$$

The payoff function is defined as:

$$\pi_i(t) = r \frac{C_i(t)}{k_i + 1} - c \cdot s_i(t)$$

where

- r is the synergy factor (public good multiplier),
- c is the cost of cooperation (set to 1),
- $k_i + 1$ is the group size (including agent i).

In a Complete Graph, where every agent interacts with all others, the payoff simplifies to:

$$\pi_i(t) = r \frac{S(t)}{N} - c \cdot s_i(t)$$

where $S(t) = \sum_{j \in A} s_j(t)$ denotes the total number of cooperators.

3.5 Evolutionary Dynamics: The Update Rule

After payoffs are computed, rational agents update their strategies according to a deterministic “copy-if-better” rule.

Each rational agent i selects a role model j for comparison:

- **On Complete Graphs:** j is chosen uniformly from $A \setminus \{i\}$.
- **On Other Graphs:** j is chosen uniformly from the neighbourhood N_i .

If $k_i = 0$, the agent has no neighbours and does not update.

The deterministic rule is:

$$s_i(t + 1) = \begin{cases} s_j(t), & \text{if } \pi_j(t) > \pi_i(t) \\ s_i(t), & \text{otherwise} \end{cases}$$

This rule replaces the stochastic Fermi updating mechanism used in standard models, corresponding to the zero-noise limit ($K \rightarrow 0$). It captures bounded rationality with perfect imitation: agents copy more successful strategies but do not experiment randomly.

3.6 Simulation Procedure

Each simulation proceeds as follows:

1. Initialization:

- Set parameters $N, f_D, r, c, \rho_R(0), T$.
- Construct the chosen network topology G .
- Assign initial strategies to agents.

2. Dynamics:

- For $t = 0, 1, \dots, T - 1$:
- Compute payoffs $\pi_i(t)$ for all agents.
- Update strategies $s_i(t + 1)$ for all $i \in A_R$ using the deterministic rule.

3. Measurement:

- After T time steps, compute the **Rational Cooperation Rate**:

$$\rho_R(T) = \frac{1}{|A_R|} \sum_{i \in A_R} s_i(T)$$

Results are averaged across 30 independent runs to reduce stochastic variation from network initialisation.

3.7 Experimental Design

Three main experiments are conducted:

1. Experiment 1 – Baseline:

- Investigates $\rho_R(T)$ as a function of f_D for $r \in \{0.5, 1.0, 1.5, 2.0, 2.5\}$.
- Topologies: Complete, Small-World, Scale-Free, Hub–Rational, Hub–Duty-Driven.
- Figure 3.1 will show cooperation outcomes across r -values.

2. Experiment 2 – Hysteresis:

- Examines dependence of the final state on the initial cooperation level $\rho_R(0)$ at fixed $r = 1.5$.
- Demonstrates lock-in and memory effects under deterministic updating.
- Figure 3.2 will display the hysteresis curves.

3. Experiment 3 – Strategic Seeding:

- Focuses on the effect of the position of committed agents in Scale-Free networks:
 - Random placement (baseline),

- Moral hubs (A_D occupy high-degree nodes),
 - Rational hubs (A_R occupy high-degree nodes).
- Figure 3.3 will compare cooperation outcomes across these cases.

3.8 Model Parameters

Table 3.1: Model Parameters

Parameter	Symbol	Value(s)	Description
Population size	N	1000	Total agents
Cost of cooperation	c	1	Individual contribution
Synergy factor	r	{0.5, 1.0, 1.5, 2.0, 2.5}	Public good multiplier
Average degree	k, m	4	Base connectivity
Duty-driven fraction	f_D	{0.1, 0.3, 0.5, 0.7, 0.9}	Fraction of committed cooperators
Time steps	T	200	Duration of simulation
Monte Carlo runs	–	30	Number of realizations
Initial rational cooperation	$\rho_R(0)$	0.5 or variable	Starting proportion of cooperators

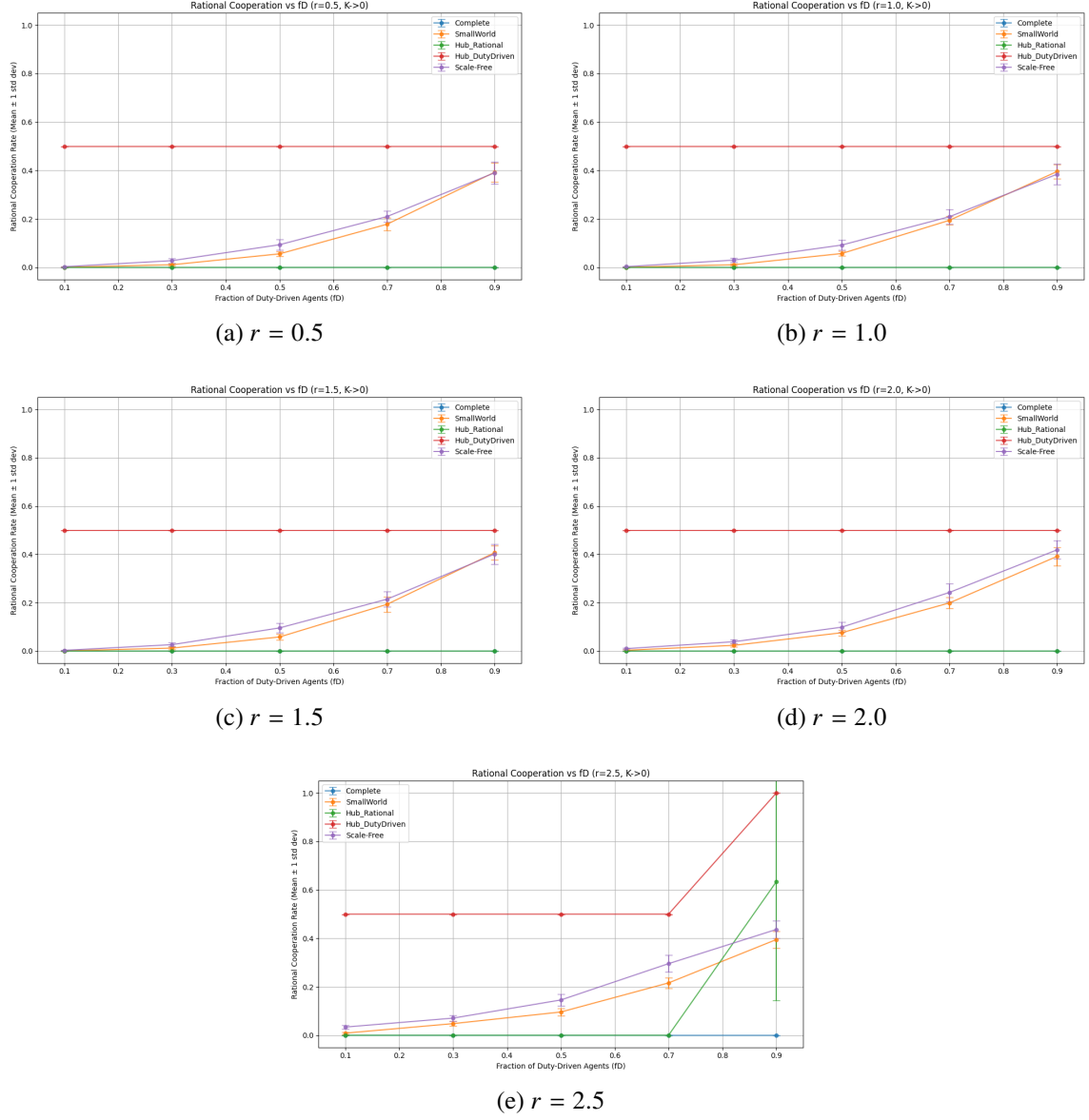


Figure 3.1: Baseline results: Final rational cooperation ($\rho_R(T)$) as a function of the duty-driven agent fraction (f_D). Each sub-plot corresponds to a different synergy factor (r). The plots compare outcomes across all five network topologies, averaged over 30 runs with $T = 200$ and $\rho_R(0) = 0.5$.

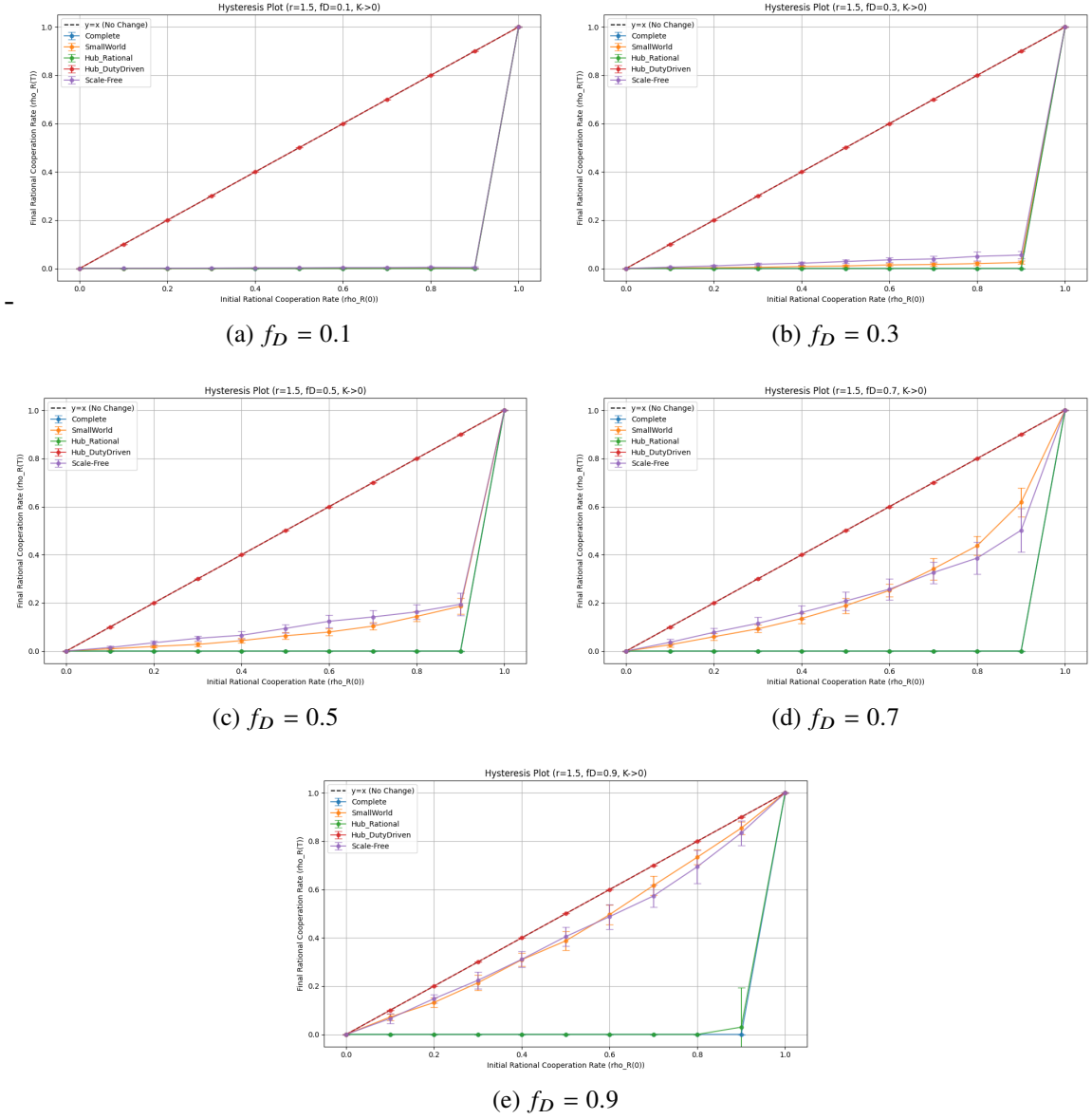
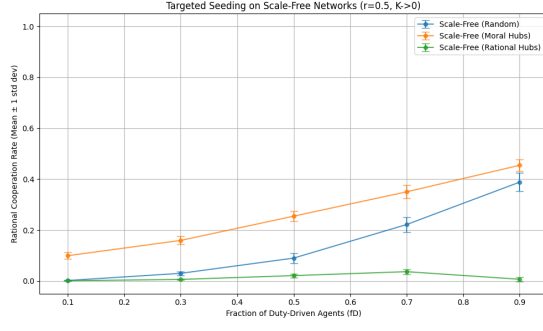
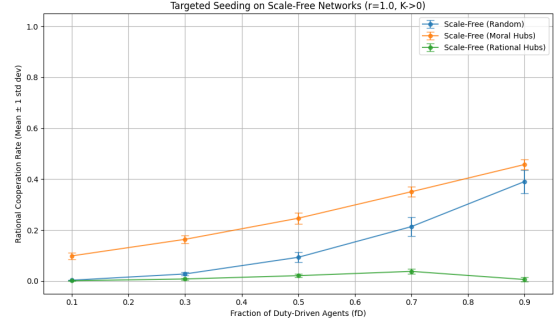


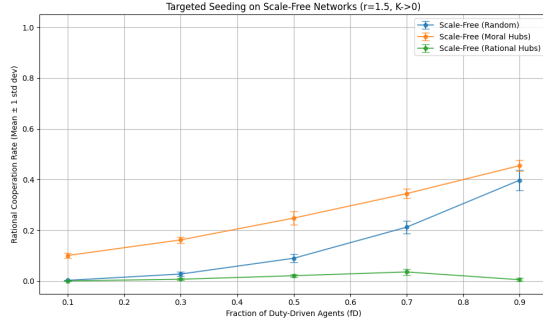
Figure 3.2: Hysteresis analysis: Final rational cooperation ($\rho_R(T)$) as a function of initial rational cooperation ($\rho_R(0)$). Each sub-plot (a-e) shows the results for a different fraction of duty-driven agents (f_D), from 0.1 to 0.9, with $r = 1.5$ fixed. The plots demonstrate that the Hub-Duty Driven topology (red line) is always "locked-in" to its initial state, while other topologies only begin to show hysteresis at higher f_D values.



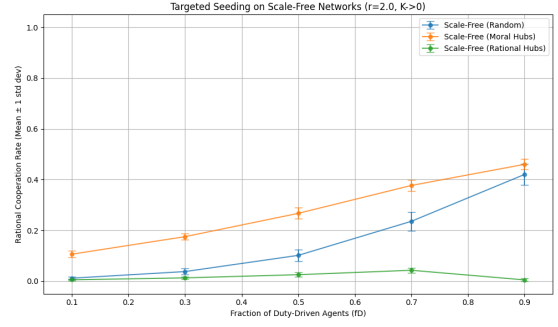
(a) $r = 0.5$



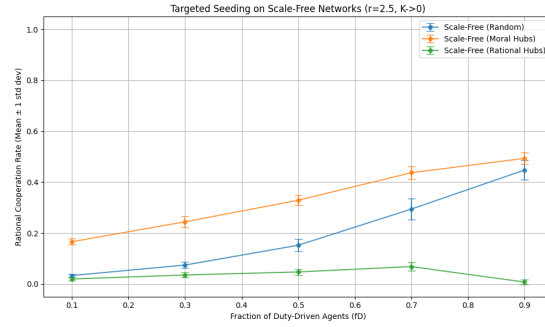
(b) $r = 1.0$



(c) $r = 1.5$



(d) $r = 2.0$



(e) $r = 2.5$

Figure 3.3: Strategic seeding in Scale-Free networks: Final rational cooperation ($\rho_R(T)$) as a function of f_D . Each sub-plot corresponds to a different synergy factor (r). The plots compare three seeding strategies: random placement (baseline), placing A_D at hubs ("Moral Hubs"), and placing A_R at hubs ("Rational Hubs").

3.9 Summary

This model bridges behavioural and structural approaches to cooperation. It captures how deterministic imitation within heterogeneous networks can produce both lock-in and cascade effects, depending on network topology and the presence of moral anchors. The next two chapters present the simulation results and interpret these outcomes through the lens of social structure and strategic positioning.

Chapter 4

Results I: The Role of Network Structure and Anchoring

In this chapter, we will examine the results from the first two experiments. This will define the baseline conditions under which cooperation can form or fail. We will look into the behaviour of the system on Complete Graphs, structured networks, and the specialised Star Graph topologies to understand the key roles of structure and moral anchoring.

4.1 The Unstructured Baseline: Failure of Cooperation

We represent the well-mixed population and random interactions of the classical model using the Complete Graph. As shown in Figure 3.1, the results are clear: for all tested values of the synergy factor r and all fractions of duty-driven agents f_D , the final rate of rational cooperation is zero ($\rho_R(T) = 0$).

This outcome supports the theoretical prediction. In a well-mixed population, rational agents face the average payoff of the entire system. Any rational defector will earn a higher payoff than a rational cooperator since they gain the shared public good $rS(t)/N$ but do not incur the cost c . Under the deterministic “copy-if-better” update

rule, this payoff advantage leads to defection spreading until all rational agents defect.

This result shows that having committed agents (A_D) alone is not enough to maintain cooperation in an unstructured environment; network structure is necessary.

4.2 Local Structure and Tipping Points

Next, we turn to the results for the Small-World and Scale-Free (Random) network topologies. Unlike the Complete Graph, Figure 3.1 indicates that these structured networks can support non-zero levels of rational cooperation.

This demonstrates that any structure is better than none. The mechanism here is local clustering. In a network, agents interact only with their local neighbourhood N_i . This allows cooperative agents (both A_R and A_D) to form clusters. Within these clusters, they are somewhat shielded from exploitation by defectors, helping them maintain a high local payoff. This finding matches existing literature on evolutionary game theory on graphs, showing that local interactions can nurture and protect cooperation.

4.3 The “Anchoring” Effect of Star Networks

Star Graph topologies provide a clear example of how the position of a single agent can influence overall results.

Hub-Rational Topology

In this setup, a single rational agent (A_R) occupies the central hub while all other agents (a mix of A_R and A_D) are peripheral leaves. This leads to a complete collapse of cooperation ($\rho_R(T) = 0$) for all parameter values. The reason is a “super-free-rider” effect. The rational hub has the highest degree ($k_i = N - 1$) and takes part in $N - 1$

local PGGs (one with each leaf). It has the greatest incentive to defect, benefitting from the contributions of any cooperating leaves. Its resulting defector payoff is the highest in the entire network. All rational leaves, whose only neighbour and role model is the hub, see this high payoff and mimic the hub’s defection. The rational hub acts like a “black hole,” actively undermining cooperation.

Hub-Duty-Driven Topology

Here, a committed agent (A_D) takes the hub’s position. The results are more complicated. At high synergy factors ($r = 2.5$), the system shows a sharp “tipping point,” moving towards near-total cooperation. This happens because the A_D hub “broadcasts” cooperation. Rational leaves compare their payoff to that of the hub. When r is high enough, a cooperating leaf’s payoff $\pi_i(\text{Coop}) = r \cdot \frac{2}{2} - c$ can surpass a defecting leaf’s payoff $\pi_i(\text{Defect}) = r \cdot \frac{1}{2}$, resulting in a wave of cooperation.

4.4 The “Strategy Lock-In”: Hysteresis Analysis

The most significant finding in this chapter comes from Experiment 2 (Hysteresis), which studies the system’s reliance on its initial state, $\rho_R(0)$. Figure 3.2 shows the results for $r = 1.5$.

The Hub-Duty-Driven topology (red line) shows perfect hysteresis: the final cooperation rate matches the initial rate ($\rho_R(T) = \rho_R(0)$). The system is completely “locked in” to its starting setup.

This result can be verified mathematically using the model’s update rule. For $r = 1.5$ and $c = 1$:

1. A rational leaf i has one neighbour: the A_D hub j . Its group size is $k_i + 1 = 2$.

2. If leaf i cooperates ($s_i = 1$), its group is $\{i, j\}$ and $C_i = 2$. Its payoff is:

$$\pi_i(\text{Coop}) = r \frac{C_i}{k_i + 1} - c \cdot s_i = 1.5 \left(\frac{2}{2} \right) - 1 = 0.5$$

3. If leaf i defects ($s_i = 0$), its group is $\{i, j\}$ and $C_i = 1$ (from the hub). Its payoff is:

$$\pi_i(\text{Defect}) = r \frac{C_i}{k_i + 1} - c \cdot s_i = 1.5 \left(\frac{1}{2} \right) - 0 = 0.75$$

4. The hub j is A_D ($s_j = 1$) and its group size is $k_j + 1 = N$. Its payoff π_j depends on how many rational leaves $\rho_R(N - 1)$ cooperate. The maximum payoff for the hub occurs when all leaves cooperate ($\rho_R = 1$):

$$\pi_j(\text{Max}) = r \frac{C_j}{k_j + 1} - c \cdot s_j = 1.5(1) - 1 = 0.5$$

5. Therefore, the hub's payoff is at most 0.5 ($\pi_j \leq 0.5$).

According to the “copy-if-better” rule $s_i(t+1) = s_j(t)$ if and only if $\pi_j(t) > \pi_i(t)$:

- A defecting leaf ($\pi_i = 0.75$) compares to the hub ($\pi_j \leq 0.5$). Since 0.5 is not greater than 0.75, the leaf never changes—it stays a defector.
- A cooperating leaf ($\pi_i = 0.5$) compares to the hub ($\pi_j \leq 0.5$). Since π_j is never strictly greater than π_i , the leaf never changes—it remains a cooperator.

The rational agents are “locked in” to their initial strategies. The committed hub acts as a form of “social memory,” perfectly preserving the initial state of the system, whether it involves cooperation or defection.

4.5 Chapter Summary

This chapter highlights two essential points. First, network structure is vital for cooperation; well-mixed populations inevitably fail. Second, the type of structure matters. A simple, uneven structure like a Star Graph can, depending on the synergy factor r , either trigger a wave of cooperation (at high r) or serve as a “social memory” that locks in the system’s initial state (at low r).

Chapter 5

Results II: Position vs. Proportion in Heterogeneous Networks

This chapter explores the main research question by examining Experiment 3: the strategic placement of committed agents within complex, heterogeneous networks. We move from simple topologies to the more realistic Scale-Free network, which is marked by a varied degree distribution, including highly connected hubs.

5.1 Central Question: Position or Proportion?

In a Scale-Free network, what is more effective for boosting cooperation across the system: increasing the proportion of committed agents (f_D) or controlling their position strategically?

To find out, we compare three scenarios:

1. **Scale-Free (Random):** The A_D agents are placed randomly. This serves as our baseline.
2. **Scale-Free (Moral Hubs):** The A_D agents are assigned to the nodes with the highest degree.

3. **Scale-Free (Rational Hubs):** The A_R agents are assigned to the highest-degree nodes, while the A_D agents are placed at the lowest-degree nodes.

The results, shown in Figure 3.1, are clear and striking.

5.2 The “Committed Anchor” Strategy (Scale-Free (Moral Hubs))

The “Moral Hubs” scenario (orange line) shows the highest level of cooperation across all values of f_D . Even a small fraction of committed agents ($f_D = 0.1$), when placed at the network’s hubs, maintains a cooperation rate that is significantly higher than in any other scenario.

The mechanism at play is *committed anchoring*. The A_D agents at the hubs are fixed cooperators ($s_i = 1$). Since hubs have the highest degree, they serve as role models for numerous other agents. They broadcast a stable cooperative strategy to a large part of the network. While a peripheral A_D agent only influences a few neighbours, an A_D hub acts as a strong, system-wide anchor that prevents defection and stabilises global cooperation.

5.3 The “Super-Free-Rider” Scenario (Scale-Free (Rational Hubs))

In contrast, the “Rational Hubs” scenario (green line) leads to a complete and immediate collapse of cooperation ($\rho_R(T) = 0$) for all $f_D < 1$. This is known as the *Super-Free-Rider effect*, which intensifies the behaviour observed in the Hub-Rational Star Graph.

A rational agent in a hub has the greatest incentive to defect. According to the

“lock-in” proof (Chapter 4), the payoff advantage for defection is:

$$\Delta\pi = c - \frac{r}{k_i + 1}.$$

With $c = 1$ and $r = 1.5$, this advantage remains positive for any agent with $k_i > 0.5$. A hub, by its nature, has a large k_i , increasing its benefit from defection.

Once the A_R hub defects, it gains a significant payoff by free-riding on all of its initially cooperating neighbours. As these neighbours evaluate their payoffs, they see the hub’s defection as the most successful strategy and follow suit. This hub acts like a “black hole,” triggering a wave of defection that spreads outward and erases all cooperation in the system.

5.4 The Baseline (Scale-Free (Random))

The random placement scenario (blue line) falls neatly between both extremes. This result aligns with expectations: it reflects the statistical average of the “Moral Hub” and “Rational Hub” cases. In any given simulation with random placement, some A_D agents will land on hubs, raising the average cooperation rate. In other cases, A_R agents will occupy those hubs, leading to system collapse. The resulting curve represents an average of these random outcomes.

5.5 Chapter Conclusion

The findings from this chapter answer the research question definitively: in heterogeneous networks, **position is much more important than proportion**.

A small number of committed agents, when placed in influential positions (hubs), can anchor and support global cooperation more effectively than a larger number of agents placed randomly. Conversely, placing self-interested, rational agents in those

same influential spots is the quickest route to systemic collapse. The structure of influence does not merely affect cooperation—it determines it.

Chapter 6

Discussion and Conclusion

In this thesis, we investigated how heterogeneous behaviour, combined with different network topologies, shapes collective action in a Public Goods Game. We showed that cooperation among agents is not only a matter of their individual preferences (always contribute or contribute only if the payoff is higher) but rather a property of the system’s architecture.

6.1 Summary of Key Findings

This thesis provides a more nuanced, structural understanding of cooperation, marking a clear progression from classical theory.

1. In line with classical arguments, we observe that in a fully mixed random population (Complete Graphs), cooperation does not emerge among agents, regardless of the presence of duty-driven agents. Hence, structure is non-negotiable.
2. Simple structures like a Star Graph create a form of *social memory*. Simple, asymmetric networks can “lock in” the initial state of the system. In the Hub–Duty-Driven Star Graph, the duty-driven hub acts as a social memory,

making the rational agents immune to strategy changes and preserving the initial level of cooperation indefinitely.

3. In complex network structures such as Scale-Free networks, an agent's position becomes the determining factor. In heterogeneous (Scale-Free) networks, position is a more powerful determinant of cooperation than the proportion of committed agents in the population.
4. Agents serve as either “Black Holes” or “Anchors.” Position has a binary effect:
 - **Committed Anchors:** When duty-driven (A_D) agents occupy network hubs, they become committed anchors that broadcast a stable cooperative strategy, preventing the system from collapsing.
 - **Super Free-Riders:** When rational (A_R) agents occupy hubs, they act as “Super Free-Riders,” using their position to maximise payoffs from defection and creating a “black hole” that destroys cooperation throughout the network.

6.2 Theoretical and Practical Implications

6.2.1 Theoretical Implications

Our findings show that agent types and network structures are not independent factors in determining cooperation. The interaction between agent type and network position emerges as a primary driver of collective outcomes. Future research should aim to identify which network structures most closely represent human social systems and how agents are positioned within them. The “Super Free-Rider” effect demonstrates that rational agents at central hubs, capable of influencing an entire population, can destabilise equilibrium in public goods contexts.

6.2.2 Practical Implications

There are clear and actionable implications for promoting cooperation in the real world. This model provides a framework for organisations seeking to enhance collective goods such as environmental activism, open-source development, or community volunteering. Identifying intrinsically motivated individuals and strategically empowering them is far more effective than attempting broad, costly interventions to “convince everyone” (i.e., increasing f_D at random). Organisations can leverage the “Committed Anchor” effect by positioning dedicated individuals in roles of influence—such as project leaders, moderators, or organisers. This *social architecture* approach utilises the existing network structure to strengthen and sustain collaboration, offering a more reliable and cost-effective pathway to cooperation.

6.3 Limitations of the Model

The simplicity of this model provides clarity but also defines its limitations.

- **Deterministic Updating:** The model assumes agents have perfect information about their neighbours’ payoffs. In reality, agents make mistakes, misperceive payoffs, or experiment with strategies (i.e., stochastic noise). Thus, the “copy-if-better” rule remains deterministic.
- **Simple Agent Types:** The population is divided into only two types, excluding “conditional cooperators,” who contribute only if their neighbours do.
- **Static Networks:** The social network is fixed. In reality, agents can rewire connections, for instance, by avoiding defectors or low-payoff partners.

6.4 Directions for Future Research

These limitations point to several promising directions for future work:

1. **Introduce Conditional Cooperators:** How would our conclusions change if agents could cooperate only when certain conditions are met?
2. **Model Dynamic Networks:** What happens if agents can modify or rewire their social connections as the game evolves?
3. **Incorporate Punishment Mechanisms:** How would the presence of exogenous rules or penalties that constrain defection affect equilibrium outcomes?

6.5 Concluding Remarks

This thesis argues that the problem of free-riding or cooperation failure is not an issue of individual morality but a collective, structural problem. The under-provision of public goods is not an unavoidable outcome, as classical theory predicts, but rather a failure of system design. By understanding human motivation and social structure, we can design societies where cooperation becomes a natural and sustainable outcome—anchored by a small number of duty-driven agents who stabilise and inspire the collective.

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